TreTechPro.ai STAR Interview Coach 3.0 Development Instructions

We are upgrading the existing STAR Interview Coach on TreTechPro.ai to Version 3.0.

I have provided three files:

1. **tretechpro_star_interview_coach_v3.json** — Interview system configuration and scoring specifications.
2. **tretechpro_star_interview_coach_v3_agent_prompt.txt** — AI interviewer and coaching instructions.
3. **tretechpro_star_interview_coach_v3_questions.csv** — Role-specific interview question bank.

Development Requirements

1. Integrate the AI Interview System

Use the JSON configuration to implement the interview workflow, evaluation framework, adaptive questioning, candidate feedback, answer retries, and progress tracking.

Validate the JSON structure and map its configuration to the existing TreTechPro.ai application.

2. Configure the AI Interview Agent

Use the provided system prompt to configure the AI interviewer.

The agent must:

- Conduct professional, realistic mock interviews.
- Ask one question at a time.
- Adapt question difficulty to the candidate's selected experience level.
- Record and preserve candidate responses.
- Evaluate answers using the appropriate scoring criteria.
- Provide a detailed evaluation after the interview.
- Offer personalized coaching and opportunities to retry answers.

3. Import the Question Bank

Import the CSV question bank into the application.

There are 17 supported career roles. Each role must have:

- 5 behavioral STAR questions.
- 5 technical knowledge questions.
- 5 technical scenario questions.

Total: 255 questions across 17 career roles.

Candidates must receive questions specific to their selected role.

4. Candidate Interview Setup

Before starting, candidates must select:

- Target career role.
- Experience level: Entry-Level, Mid-Level, or Senior.
- Interview type: Full Interview, Behavioral, Technical Knowledge, or Technical Scenarios.

The full interview contains 15 questions. Individual practice categories contain 5 questions.

5. Candidate Response Recording

Candidates must be able to answer using voice or text.

For voice responses, implement audio recording and transcription.

Save each response with its corresponding question, interview session, selected role, and experience level.

6. Interview Evaluation

Behavioral responses must be evaluated using the STAR framework:

- Situation: 20%
- Task: 20%
- Action: 40%
- Result: 20%

Technical knowledge responses must be evaluated on technical accuracy, completeness, reasoning, and communication.

Technical scenario responses must be evaluated on problem identification, proposed actions, validation, and communication.

Provide individual question scores, category scores, and an overall interview score.

7. AI Coaching and Answer Improvement

After the interview, the AI must:

- Identify candidate strengths.
- Explain weaknesses in their responses.
- Identify missing STAR components.
- Explain technical knowledge gaps.
- Demonstrate how to structure stronger answers.
- Allow candidates to retry individual questions.
- Compare original and improved responses.

The AI must not invent accomplishments, experiences, or technical results on behalf of the candidate.

8. Candidate Dashboard

Create a dashboard where candidates can:

- Review completed interviews.
- Access original answers and evaluations.
- Review their STAR and technical scores.
- Track improvement across interview sessions.
- Retry questions.
- View personalized practice recommendations.
- Download their interview evaluation reports.

9. Adaptive Interviewing

The AI must adjust technical difficulty and evaluation expectations based on the candidate's selected experience level.

The system should also support relevant follow-up questions when a candidate's response needs clarification.

Follow-up questions must not replace the required 15 primary interview questions.

10. Testing and Validation

Before deployment, verify:

- All 17 career roles load correctly.
- Each role has exactly 15 primary questions.
- Behavioral questions receive STAR evaluations.
- Technical questions receive technical evaluations.
- Candidate recordings and transcripts are saved correctly.
- The AI produces consistent structured scoring results.
- Retried answers do not overwrite original answers.
- Interview history and progress tracking work correctly.
- Reports display the correct scores and coaching feedback.

Please integrate these capabilities into the existing TreTechPro.ai STAR Interview Coach while maintaining the current website's branding and user experience.

Primary objective: Create a realistic AI-powered interview simulator that teaches candidates how to structure their answers, communicate technical expertise, demonstrate problem-solving skills, and improve their performance in actual job interviews.